

# Correct before the deadline - or NONE?

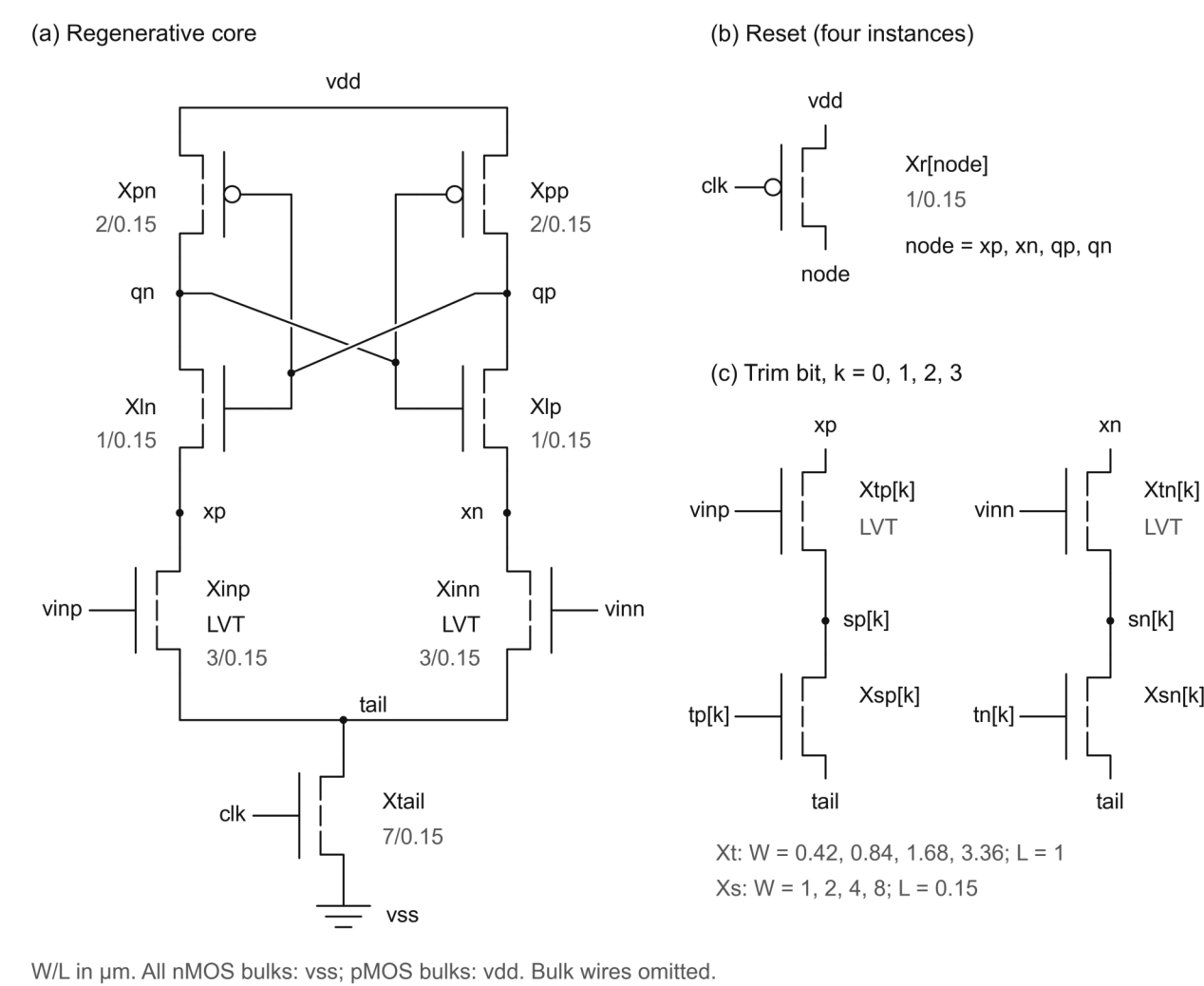
Comparator Atlas | specify, qualify, then compare energy

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## What a choice must satisfy

Calibration alone is not qualification.  
Every included point must be correct on time.



Recorded circuit topology. Nine declared sizing candidates; the lower-energy control was evaluated after selection.

## Correct, wrong or unresolved

A correct decision reaches complementary 80% / 20% output rails with the expected polarity before the deadline. Wrong polarity and unresolved outputs are counted separately.

Calibration and full-cycle core energy are evaluated from archived measurements. A later readout does not rerun SPICE or shorten the measured energy cycle.

## Specify, qualify, then choose

Calibrated SCHEMATIC: local\_boundary; all 49 controlled-width-stress conditions. Both signs at finite sampled inputs through 30 mV; 1 ns deadline.

**$\geq 1$  mV NONE**

No feasible compared design

Mean energy / limits: not defined

**$\geq 3$  mV Selected: lvt\_balanced\_4b**

294/294 correct at all included samples

249.655 fJ mean core / cycle

**$\geq 30$  mV Control: lvt\_base\_3b**

98/98 correct at all included samples

150.531 fJ mean core / cycle

Post-hoc: qualify, then rank three designs. Different-band energies are not a same-spec gain. No interpolation or global optimum.

**Selected  $\geq 3$  mV band / same 294 samples**

**154.517 ps**

**363.217 fJ**

Min. deadline - recorded decision time (sampled)

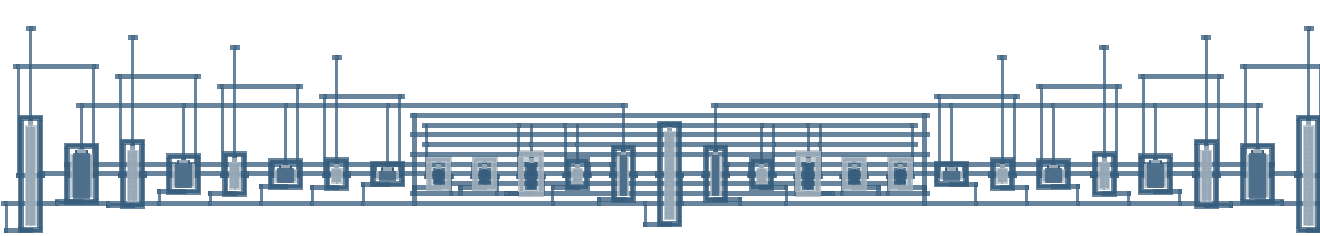
Maximum sampled core energy

**20 wrong samples remain at 2 ns**

Exactly the same keyed set at 1 ns and 2 ns, all at signed +/-1 mV. Locations and trim codes do not identify a physical failure cause.

## Separate matched layout example

Local-pitch b1 vs adopted a1; known nonblind, code-zero: 45 PVT conditions x four signed inputs (-10, -3, +3, +10 mV). 180 keys per layout / C or RC; ngspice-47.

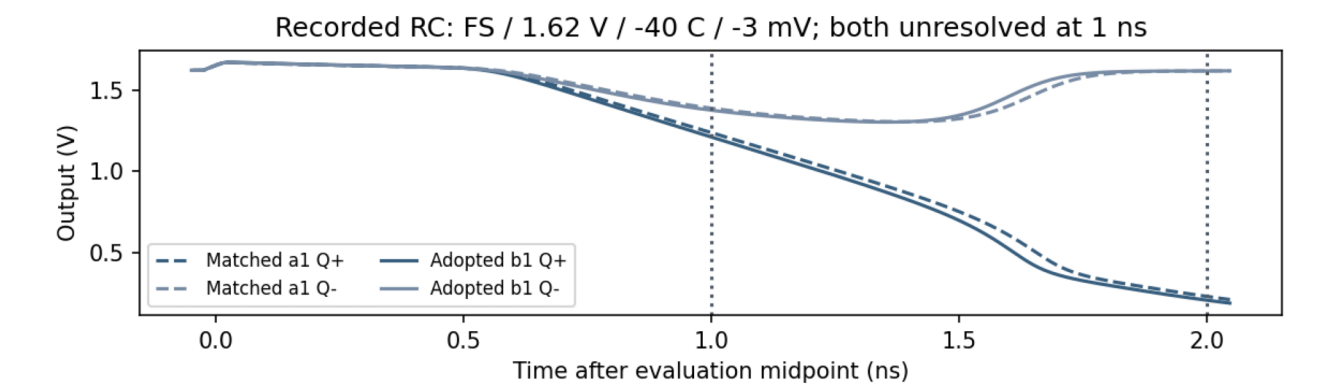


Bbox (um2): adopted 1873.300 / matched a1 2074.254 (9.69% less).  
Historical native proofs reused, not rerun; no signoff.

| Correct / 180 | 1 ns    | 2 ns    |
|---------------|---------|---------|
| Both C        | 170/180 | 180/180 |
| a1 RC         | 156/180 | 180/180 |
| b1 RC         | 158/180 | 180/180 |

RC mean: 421.04 -> 412.88 fJ / cycle

1 ns RC unresolved: a1 24 -> b1 22; zero wrong.  
Only SS / 1.62 V / 125 C / +/-10 mV improve; FS remains late.  
2 ns primary, prospectively declared; no lost correct keys.



Matched a1 -> b1; worst RC: 1.811 -> 1.760 ns.  
~1.94% energy change is not robust global PPA.

**Archived RC-deck outcomes; model physical fidelity not yet qualified.**

[Source Notebook](#)

[Source Colab](#)

No qualified PEX or silicon prediction. C-only is not independent ground truth; RC-versus-C differences do not isolate resistance.

Finite samples, not foundry Monte Carlo/yield, noise/jitter/PVT confidence, timing signoff or a worst-cycle guarantee. Core excludes drivers and calibration/controller infrastructure.

References: Razavi (2015), DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka (2022), DOI 10.1007/s10470-022-01992-6.

Original code: MIT; SKY130: Apache-2.0. GitHub Copilot assisted implementation, experiment automation, figures and documentation; the author is responsible.